

Salt Tectonics, Halokinesis & Diapiric Piercement Folds

Authoritative Reference: Ben A. van der Pluijm & Stephen Marshak, *Earth Structure* (2nd Ed., Chapters 2 & 10: Salt Structures & Special Fold Geometries)

Comprehensive Chapter Masterclass: Halokinesis & Diapiric Folding

Rheology of Rock Salt, Gravitational Instability, Piercement Mechanics & Diapiric Folds

1. Physical Properties of Rock Salt & Halokinesis

Halokinesis (salt tectonics) refers to the autonomous, gravity-driven flow and intrusion of rock salt (predominantly halite, NaCl) beneath sedimentary basins. The extraordinary ability of rock salt to flow and intrude overlying rock sequences is governed by three fundamental physical properties (Van der Pluijm & Marshak, Chapter 2):

- Incompressible Low Density:** Halite has a fixed density of $\sim 2.16 \text{ g/cm}^3$ that remains nearly constant regardless of burial depth. In contrast, common terrigenous sediments (muds, shales, sands) deposit at lower densities ($1.8\text{--}2.0 \text{ g/cm}^3$) but undergo irreversible mechanical compaction, exceeding $2.2\text{--}2.6 \text{ g/cm}^3$ below ~ 1 km of burial. This establishes a profound **Rayleigh-Taylor gravitational density inversion** where a buoyant, low-density salt layer is trapped beneath a heavier overburden.
- Extremely Low Yield Strength:** Salt behaves as a non-Newtonian fluid under geologic strain rates, undergoing pervasive crystal-plastic dislocation creep at temperatures as low as 50°C – 100°C .
- Differential Loading:** Tectonic extension, lateral sediment thickness variations, or prograding sedimentary wedges create pressure gradients that trigger buoyant upward ascent of mobile salt.

*"Table 2.5 — Terminology of Salt Structures: **Salt Diapir:** A salt structure that pierces bedding in overlying strata... In a diapir fold (or piercement fold), ductile, incompetent material (such as rock salt, gypsum, or overpressured mud) has broken through the more brittle overlying rock layers and occupied the core of the fold."*

— Van der Pluijm & Marshak, *Earth Structure* (2nd Ed., Table 2.5, p. 28)

Salt stock

Equant salt diapir in plan view. Mature salt diapirs are generally narrower at depth and broader at the top; the lower part is called the *stem* and the upper part is called the *bulb*.

Salt wall

Elongate salt diapir in plan view.

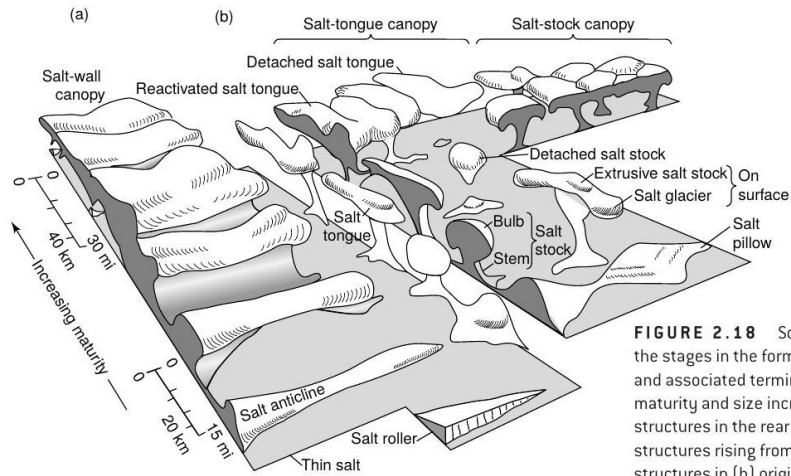


FIGURE 2.18 Schematic diagram showing the stages in the formation of salt structures and associated terminology. Structural maturity and size increase toward the structures in the rear. Sequence in (a) shows structures rising from line sources, whereas structures in (b) originate from point sources.

Figure from Van der Pluijm & Marshak (Chapter 2): Schematic cross section of a salt diapir (diapir fold). The buoyant, ductile salt core has sheared and pierced through the brittle overlying strata, dragging and upturning the punctured beds along its margins.



FIGURE 10.18 Monocline in the Big Horn Mountains (Wyoming, USA).

10.5 SOME SPECIAL FOLD GEOMETRIES

We end the descriptive part of this chapter with a few special fold geometries that you may encounter in your field career. **Monoclines** are fold structures with only one tilted limb; the beds on either side of the tilted limb are horizontal. Monoclines typically result from a vertical offset in the subsurface near the tilted portion of the structure. The fault uplifts a block of relatively rigid igneous or metamorphic rock, and the overlying sedimentary layers drape over the edge of the uplifted block to form the monocline (Figure 10.18). Spectacular examples are found in the Colorado Plateau of the western United States. **Kink folds** are small folds (less than a meter) that are characterized by straight limbs and sharp hinges. Typically they occur in finely laminated (that is, strongly anisotropic) rocks, such as shales and slates (Figure 10.19). Sharply bending a deck of cards is a good analogy for the kinking process, because kink folds are formed by displacements between individual laminae (individual cards in the analogy). **Chevron folds** (Figure 10.8) are the larger-scale equivalent of kink folds.

The term **box fold** describes a geometry that is pretty self-explanatory (Figure 10.5). In order for a box fold to form, a layer must be detached from the underlying and overlying layers. They are therefore common in areas with weak basal layers, such as in the Jura Mountains of Switzerland. **Ptygmatic folds** are irregular and isolated



FIGURE 10.19 Kink folds in mica-rich portion of greywackes of the Cantabrian Mountains (northern Spain).

Evolutionary Stages of Salt Structures: From (1) an initial concordant salt pillow / dome, to (2) an intrusive piercement diapir (diapir fold), to (3) allochthonous salt sheets, canopies, and subaerial salt glaciers (namakiers).

2. The Piercement Mechanism & Fold Core Occupation

When an anticlinal fold develops above an ascending salt ridge, the ductile core material exerts strong tensile and upward shear forces on the overlying rocks:

- **Concordant Stage (Salt Dome / Pillow):** The overlying strata bend upward smoothly in conformity with the salt swell without breaking.
- **Piercement Stage (Diapir Fold / Piercement Fold):** The stretching capacity of the brittle roof is exceeded. Faulting and tensile rupture occur, allowing the mobile salt plug to punch through (pierce) the brittle carapace and occupy the central core of the fold. Strata on the flanks are dragged into subvertical, overturned, or faulted attitudes adjacent to the diapir walls.



Official Exam Question Statement & In-Depth Critical Analysis

Directorate General of Mines Safety (DGMS) 2019 · Junior Scientific Officer Exam · Question 14

Question Statement

Q.14: In a fold, rock salt has broken through the more brittle overlying rocks and occupied its core. The structure is known as:

(a) Disharmonic fold

(b) Drag fold

(c) Box fold

(d) Diapir fold

Correct Answer: (d) Diapir fold



A fold in which mobile, ductile material (such as rock salt) has actively pierced through the more brittle overlying rock carapace to occupy the core is defined as a diapir fold (or piercement fold).

Exhaustive Option-by-Option Critical Analysis

(d) Diapir fold

CORRECT

Why it is uniquely correct: In structural geology and salt tectonics (Van der Pluijm & Marshak, 2004, Table 2.5, p. 28; Billings, 1972, p. 101; Davis & Reynolds, 1996), a **diapir fold** (also known as a *piercement fold* or *diapiric fold*, from the Greek *diapairein* = to pierce through) is explicitly defined as a fold whose core is composed of mobile, incompetent material (such as rock salt, gypsum, or shale) that has ruptured and broken through the more brittle overlying rock layers.

(a) Disharmonic fold

INCORRECT

Why it is incorrect: A *disharmonic fold* refers to a layered sequence where adjacent layers exhibit markedly different fold profiles, wavelengths, or amplitudes due to sharp contrasts in rheological competence (e.g., incompetent shale folding disharmonically between competent sandstone beds). In disharmonic folding, layers remain stratigraphically intact and do **not** pierce or punch through each other.

(b) Drag fold

INCORRECT

Why it is incorrect: A *drag fold* is a minor, asymmetric parasitic fold formed in incompetent beds adjacent to a fault plane due to frictional dragging of the wall rocks during fault displacement, or by interlayer shear during flexural-slip folding. Drag folds do not involve diapiric piercement of rock salt through a brittle roof.

(c) Box fold

INCORRECT

Why it is incorrect: A *box fold* (conjugate fold) is a profile geometry characterized by two distinct hinge lines and axial planes that converge downward, creating a broad, flat-topped crest and steep limbs. It is a geometric fold shape, not an intrusive piercement structure.



Synthesis Reference Tables, Diagnostic Criteria & High-Yield Practice

Salt Structure Classification & 8 Exam-Grade Multiple-Choice Questions

Classification of Salt Structures (Van der Pluijm & Marshak, Table 2.5)

Structure Name	Relationship to Overlying Bedding	Geometric Form	Tectonic Stage
Salt Pillow	Concordant (overlying strata conform to the bulge without rupture)	Sub-circular to elliptical upward swell	Immature early buoyant uplift
Salt Roller	Asymmetric, faulted concordant/sub-concordant	Low-amplitude triangular salt wall in extensional fault footwalls	Syn-rift extension
Salt Diapir (Diapir Fold)	Discordant / Piercing (breaks through overlying strata and occupies core)	Cylindrical plug, chimney, or intrusive wall	Mature piercement stage
Salt Canopy	Discordant feeder spreading horizontally	Coalesced sheet of multiple expanding diapir heads	Advanced extrusion/spreading

8 High-Yield Practice Questions on Salt Tectonics & Fold Terminology

1. What physical phenomenon causes rock salt to become buoyant relative to overlying sediments below depths of approximately 1 km?

- (a) Salt density increases dramatically
- (b) Sediments undergo compaction increasing their density above that of salt ($\sim 2.16 \text{ g/cm}^3$)
- (c) Salt vaporizes into steam
- (d) Magnetic repulsion

Answer: (b) — Salt does not compact and retains a constant density of $\sim 2.16 \text{ g/cm}^3$, whereas sediments compact to $>2.4 \text{ g/cm}^3$, creating a density inversion.

2. The term "piercement fold" is an exact synonym for:

(a) Diapir fold (b) Chevron fold (c) Ptygmatic fold (d) Isoclinal fold

Answer: (a) Diapir fold — "Diapir" is derived from the Greek word meaning "to pierce through."

3. A salt structure that bulges upward concordantly without rupturing or piercing the overlying strata is called a:

(a) Salt diapir (b) Salt pillow (or salt dome) (c) Salt canopy (d) Salt glacier

Answer: (b) Salt pillow (or salt dome) — In a pillow or dome, the overlying beds remain concordant and continuous.

4. In petroleum geology, why are salt diapirs world-class structural traps for hydrocarbons?

- (a) Salt is porous and stores large volumes of oil
- (b) Impermeable salt seals truncate upturned, porous reservoir beds on the flanks of the diapir
- (c) Salt generates oil by chemical reactions
- (d) Salt creates volcanic eruptions

Answer: (b) — Crystalline halite has virtually zero permeability, acting as an exceptional top and lateral seal against upturned reservoir strata.

5. Subaerial extrusion of salt that flows downslope under its own weight in arid environments (such as the Zagros Mountains of Iran) is termed a:

(a) Namakier (salt glacier) (b) Moraine (c) Esker (d) Drumlin

Answer: (a) Namakier (salt glacier) — Active salt extrusions flowing like glaciers are called namakiers.

6. Folds that develop different geometric forms and wavelengths in adjacent layers due to competence differences are called:

(a) Harmonic folds (b) Disharmonic folds (c) Diapir folds (d) Supratenuous folds

Answer: (b) Disharmonic folds — Differences in layer stiffness produce disharmonic folding.

7. The cap rock commonly found capping salt diapir heads consists of an insoluble residue dominated by:

(a) Anhydrite, Gypsum, and Calcite (b) Pure Quartz (c) Olivine (d) Diamond

Answer: (a) Anhydrite, Gypsum, and Calcite — Groundwaters dissolve halite, leaving behind a residual crust of insoluble sulfates and secondary carbonates.

8. What is the dominant crystal deformation mechanism that enables halite to flow ductily at geologic strain rates?

- (a) Dislocation creep and pressure solution (b) Melting (c) Sublimation (d) Explosive shattering

Answer: (a) Dislocation creep and pressure solution — Halite exhibits low activation energy for dislocation glide/climb and solution-precipitation creep.